Brief and AI Co-editing Paper on Geologist-GPT and Procedural Game Terrain Generation

*This is a **sub-entry of game-related competition** of how AI kicks in indirectly; I hope to get some inspiration from this paper for future projects.

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Part IV, V co-editing: Grok-3 & o4-mini Generative AI

I. Background

I participated in Hacktoberfest 2023 in October 2023. Participants should upload 4 public pull requests to complete this year's requirements. One of the submission topics for this event is Prompt Engineering, and I desire the topic to be mainly related to game content planning, design, and architecture.

II. My thoughts on Geology

Terrain creation or generation is one of the game design topics, using different aspects of knowledge such as geography, culture, etc, to make stunning scenery, and proper implementation of lore, being a part of the backbone for your game. This essay uses "Geology as science" and "Cultural as art" as a base to create something on top of it.

III. Topic investigation in different sectors

Topic investigation

"Geology" is one of the topics in "Geography", and it is typically used to **study the Earth as a science subject**. The significant question is about the constitution of Earth, and how the Earth use this composition to form different biomes, terrains, climates, or even biodiversities, etc I think the topic can not stop at the scientific level, but art, humanly, liberal...etc...

In human geography

Further, this interdisciplinary approach also affected human geography **as an art subject contributor**. The same species of humans live in certain biomes, and terrains may have a variety of different behaviours.

Firstly, the **materials** to build houses in grass plains and mountainous areas may be diverse. Grassland huts are constructed mainly with straw and dry dirt bricks, but alpine huts with rocks and raw wood, which are environmentally easy to access. The material used is also impacted by real-world physics and engineering problems.

Moreover, people would also take raw materials in their living biomes to craft items to contribute to their cultures, no matter religion, production tools, entertainment, etc Specifically, wood and stone to carve their gods, totems, and beliefs; clay and

Environmental factors such as temperature, humidity, and micro-climate are also heavily considered factors **in relation to materials selection, let** we will explain in the later part.

Secondly, the **land use** in different biomes is also diverse. In the steep areas, they use a "ladder field", which spreads across the field vertically down (or up) vertically on the mountain. By using gravity to transport water from the top, via a water channel, to irrigate most areas of the field.

Also, they built **structures** to keep freshwater fishes and animals (like edible frogs, and clams...) in the field, fish enhanced the field ecosystem in the field because they ate up withering away crops and fertilised fields by pooing. It is a concept that is proven in modern technology, named "Aquaponics". How clever ancient people are!

IV. AI (Grok-3) co-editing

Geology, a subset of geography, is the scientific study of Earth's composition, structure, and processes. It addresses fundamental questions about how Earth's materials and forces shape biomes, terrains, climates, and biodiversity. However, I believe geology's relevance extends beyond science into the realms of art, humanities, and cultural studies. By combining these perspectives, we can create richer, more meaningful virtual environments in games.

In the context of procedural terrain generation, geology provides a framework for simulating realistic landforms, while cultural insights ensure that these environments resonate with human experiences and narratives. This interdisciplinary approach can enhance both the aesthetic and functional aspects of game worlds.

Topic Investigation in Different Sectors

Topic Investigation: Geology as a Foundation

Geology seeks to understand Earth's physical makeup and the processes that form diverse landscapes. This knowledge is crucial for procedural terrain generation in games, as it enables the creation of believable environments—mountains shaped by tectonic activity, river valleys carved by erosion, or deserts formed by arid climates. However, limiting geology to a purely scientific lens overlooks its potential to inspire artistic and cultural depth in game design.

In Human Geography: The Cultural Lens

Human geography examines how environments influence human behaviour, culture, and societal development. This perspective is vital for creating game worlds that feel lived-in and authentic. Below are key ways in which terrain and biomes shape human activity and culture, which can be reflected in procedural terrain generation:

1. Material Use and Architecture:
   The materials available in a biome heavily influence the construction of shelters and tools. For instance, grassland communities might build huts from straw and mud bricks, while alpine dwellers use rocks and raw wood—materials abundant in

their respective environments. These choices are also dictated by real-world physics and engineering constraints, such as the need for insulation in cold climates or lightweight structures in windy plains. In game design, procedural algorithms can simulate these material preferences based on terrain type, adding realism to settlements.

Additionally, environmental factors like temperature, humidity, and microclimates affect material selection. For example, tropical regions might favour bamboo for its flexibility and resistance to moisture, while arid regions rely on stone for durability. Procedurally generated villages in games can reflect these nuances by dynamically selecting building materials based on simulated climate data.

2. Cultural Artefacts and Practices:
   Humans often use local resources to craft items central to their culture, including religious symbols, tools, and entertainment objects. Wood and stone might be carved into totems or deities, while clay is shaped into pottery. These cultural elements can be procedurally integrated into game environments, with terrain type dictating the availability of resources and, consequently, the cultural artefacts present in a region.

3. Land Use and Agricultural Practices:
   Terrain shapes how humans utilise land. In mountainous regions, ancient societies developed terraced fields (or "ladder fields") to cultivate crops on steep slopes. Water channels harnessed gravity to irrigate these fields, often supporting integrated ecosystems like aquaponics, where fish consumed decaying plants and fertilised the soil with their waste. Such ingenuity can inspire procedural generation systems to create terraced landscapes with functional water systems and dynamic ecosystems, enhancing both realism and gameplay potential (e.g., farming mechanics).

In the Scientific Aspects of Geography

From a scientific standpoint, terrain formation is driven by geological processes influenced by climate, tectonics, and erosion. For instance:

- Plate tectonics creates mountain ranges and rift valleys.
- Weathering and erosion shape riverbeds and canyons.
- Climate determines vegetation and soil types, such as lush forests in temperate zones or sparse shrubs in deserts.

Procedural terrain generation can simulate these processes using algorithms like Perlin noise for natural randomness, combined with rules mimicking geological forces. For example, a game could dynamically generate mountain ranges with realistic fault lines or river systems that follow gravitational paths, ensuring that the terrain feels organic and scientifically plausible.

Geologist-GPT: A Tool for Procedural Terrain Generation

To bridge the scientific and cultural dimensions of terrain generation, I propose the concept of "Geologist-GPT"—a specialized AI model tailored for game design. This tool would combine geological data with cultural insights to assist developers in creating procedurally generated terrains that are both realistic and narratively rich.

1. Scientific Modeling:
   Geologist-GPT can simulate geological processes by drawing on datasets of Earth’s landforms, climate patterns, and biomes. It could generate heightmaps, erosion patterns, and vegetation distribution based on scientific principles, ensuring that terrains adhere to natural laws.

2. Cultural Integration:
   The model can also incorporate human geography data, suggesting cultural elements based on terrain type. For example, in a procedurally generated desert, Geologist-GPT might propose nomadic settlements with tents made of animal hides, reflecting resource scarcity and mobility. In a mountainous region, it could suggest cliffside villages with stone architecture and terraced farming.

3. Prompt Engineering for Creativity:
   As part of Hacktoberfest's focus on prompt engineering, developers can use Geologist-GPT with tailored prompts like:

   - "Generate a tropical island with volcanic terrain and a fishing-based tribal culture."
   - "Create a tundra biome with glacial features and nomadic herding communities."
     Such prompts would guide the AI to balance geological accuracy with cultural storytelling, producing unique and immersive game worlds.

Material Selection in Procedural Generation

Material selection is a critical aspect of terrain generation, influencing both the visual style and gameplay mechanics of a game. Procedurally, material distribution can be tied to biome and climate data:

- Rock and Soil Types: Mountains might feature granite or basalt, while river valleys have fertile alluvial soil. These materials can affect gameplay, such as mining yields or farming productivity.
- Vegetation and Resources: Forests provide timber and wildlife, while deserts offer sparse cacti and rare minerals. Procedural systems can allocate resources based on terrain, encouraging players to adapt to environmental constraints.
- Cultural Artifacts: As discussed, materials shape cultural output. A procedural system could generate unique tools, clothing, or architecture based on available resources, adding depth to NPC interactions and world-building.

Conclusion

Procedural terrain generation in games stands to benefit immensely from an interdisciplinary approach that combines geology as science with culture as art. By simulating natural processes and human responses to environments, developers can create virtual worlds that are both believable and engaging. The proposed Geologist-GPT tool exemplifies how AI and prompt engineering can facilitate this process, offering a scalable solution for generating terrains rich in both detail and narrative.

Through my participation in Hacktoberfest 2024, I aim to contribute to the field of game design by exploring innovative ways to merge scientific accuracy with cultural depth. Future work could involve developing open-source algorithms for Geologist-GPT or creating sample terrains as proof of concept, fostering collaboration within the game development community.

V. AI (o4) co-editing

Background and Related Work

Procedural Terrain Generation

- **Noise-Based Methods** (Perlin 1985; Simplex noise): Fast, random yet coherent heightfields.
- **Hydraulic and Thermal Erosion** (Smelik et al. 2009): Simulate water flow and temperature cycles to carve realistic valleys and rock faces.
- **Biome Blending** (Kelly et al. 2018): Layer vegetation, soil, and climate zones to create diverse ecosystems.

Geology & Human Geography

- **Plate Tectonics**: Drives mountain building (orogeny), rift valleys, and continental shapes.
- **Rock Cycle**: Igneous, sedimentary, and metamorphic processes determine soil fertility, mineral availability, and landform stability.
- **Cultural Adaptation**: Human settlements reflect resource availability (stone, timber, clay), climate constraints (humidity, temperature), and agricultural techniques (terracing, aquaponics).

The Geologist-GPT Framework

Model Architecture

- **Base LLM**: Fine-tuned on geology textbooks, cultural anthropology studies, and historical land-use records.
- **Knowledge Modules**:
  - *Lithology & Stratigraphy*: Rock types, mineralogy, soil profiles.
  - *Geomorphology*: Erosion rates, sediment transport, floodplain dynamics.
  - *Ethnogeography*: Material culture, vernacular architecture, traditional agriculture.

Prompt Engineering for Terrain Pipelines

Designers or procedural engines issue prompts such as:

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Generate a rugged mountain range composed of metamorphic schist with
southern-facing terraced vineyards fed by gravity-fed aqueducts. Include
local hamlets using raw wood and slate roofing.
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Geologist-GPT returns:

- Ideal slope gradients (30–45° for schist stability)
- Terrace spacing (~1–2 m steps with 0.5 m rise)
- Aqueduct channel dimensions (cross-section 0.2 m^2 for 2 L/s flow)
- Local tree species (Pinus sylvestris) and woodworking styles
- Settlement layouts (hamlet of 20–30 houses clustered near water source)

Case Studies

Grassland Plains Village

Prompt:

> "Design a temperate grassland village on loess soils with seasonal rainfall."

Geologist-GPT Output:

- Soil: 1.5–2 m deep loess—ideal for rammed-earth bricks and thatch huts.
- Building Materials: Straw bale reinforcement; "Cob" walls with 30% clay content.
- Climate Adaptations: Raised plinths (30 cm) to avoid seasonal flooding.
- Agriculture: Rotational plots of wheat, pulses, and root vegetables; buffer strips of willows for erosion control.

Integration: Procedural engine uses these parameters to:

1. Modulate noise parameters for gentle rolling hills (slope < 10°).
2. Place clusters of huts with mud-cob textures.
3. Generate field boundaries with hedgerow meshes.

Alpine Aquaponics Terrace

Prompt:

"Create steep terraced fields at 1 500 m elevation, integrating aquaponics."

Geologist-GPT Output:

- Terrace geometry: 2 m wide benches, 0.8 m risers (stone retaining walls).
- Water Source: Meltwater channels insulated with timber flumes to maintain 4–12 °C.
- Aquatic Fauna: Cold-water trout species; edible Rana temporaria (frogs).
- Farming Cycle: Spring planting of barley and cold-tolerant vegetables; fish harvest in autumn.

Integration:

- Engine carves vertical noise profiles for step-like landforms.
- Assigns stone and timber shaders to walls and flumes.
- Populates fishpond meshes with particle effects (ripples, foam).

Discussion

1. **Authenticity vs. Performance**: Injecting scientific detail increases computation and asset variety. Mitigation: cache common patterns (e.g., standard terrace modules).
2. **Lore Coherence**: Geologist-GPT can weave local myths (mountain spirits tied to schist cliffs) into procedural place names and decorations.
3. **Extensibility**: Future modules could cover planetary geology (lava plains on exoplanets) or deep-sea terrains.

Conclusion

By uniting the rigor of geology with the richness of cultural geography, **Geologist-GPT** elevates procedural terrain generation from mere noise functions to world-building with scientific and anthropological fidelity. This approach nurtures more immersive environments, fosters storytellers' creativity, and lays a robust foundation for next-generation game worlds.